Advanced Computer-Aided VLSI System Design

Phase-Locked Loop (PLL)

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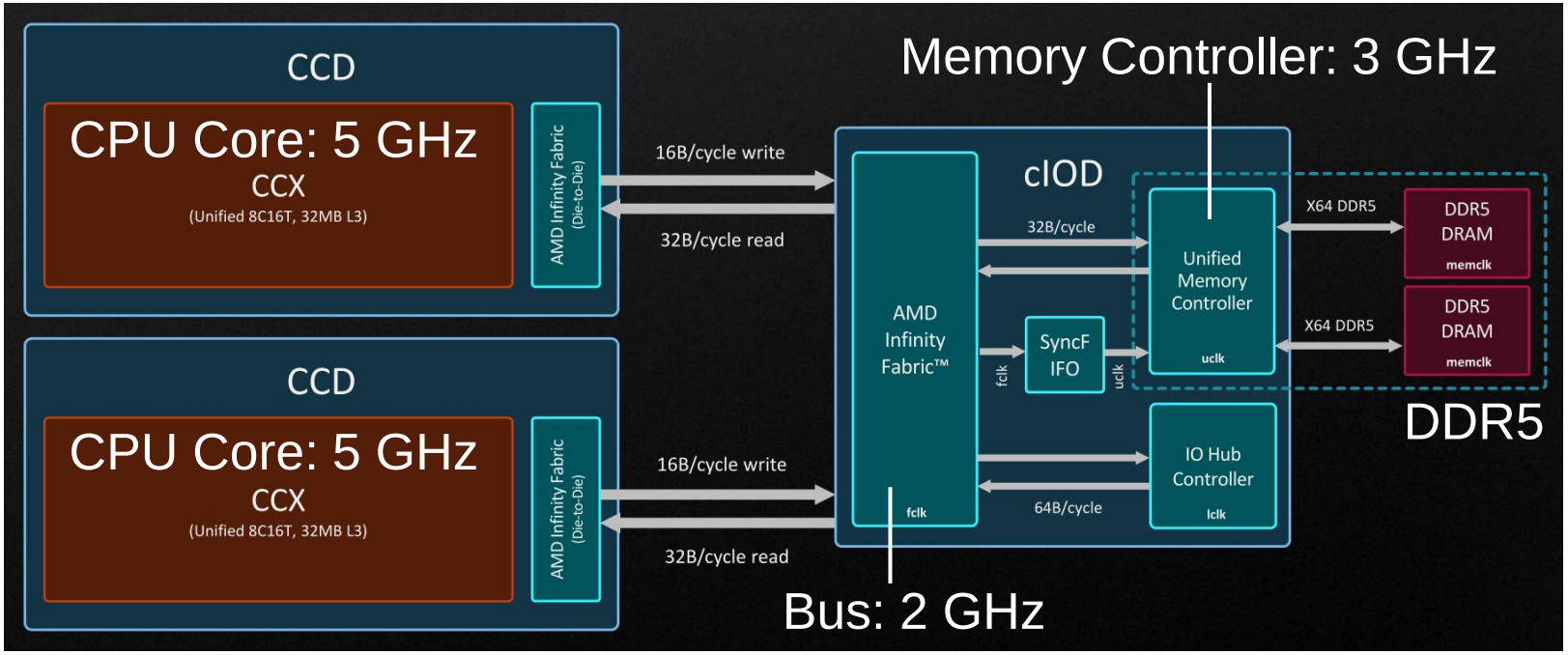


Phase-Locked Loop

- A phase-locked loop (PLL) is widely used in various electronic systems
- Clock Generation
  - Generates high-frequency clocks from a low-frequency reference
- Frequency Synthesis
  - Used in radio transmitters and receivers to generate stable and precise frequencies
- Synchronization
  - Aligns clock and data in high-speed serial transmission, e.g., clock and data recovery (CDR)

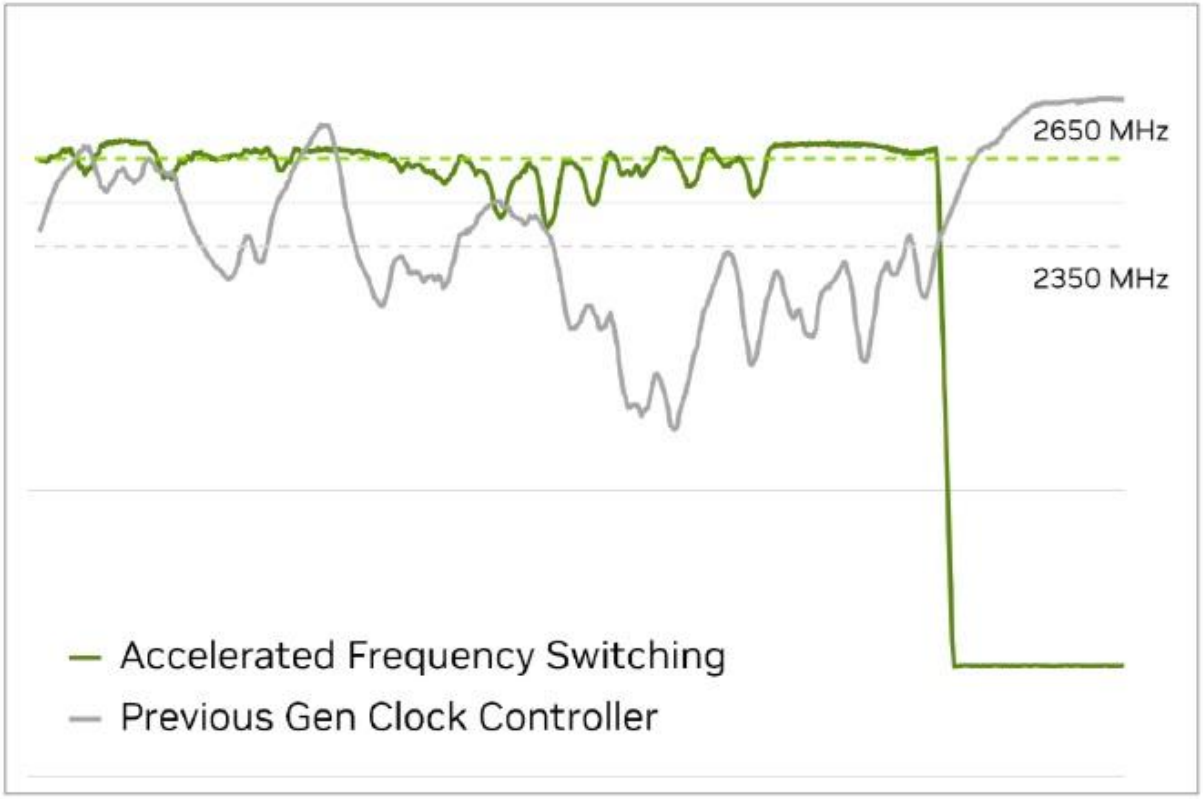
Product Examples

- AMD Zen 5 CPU



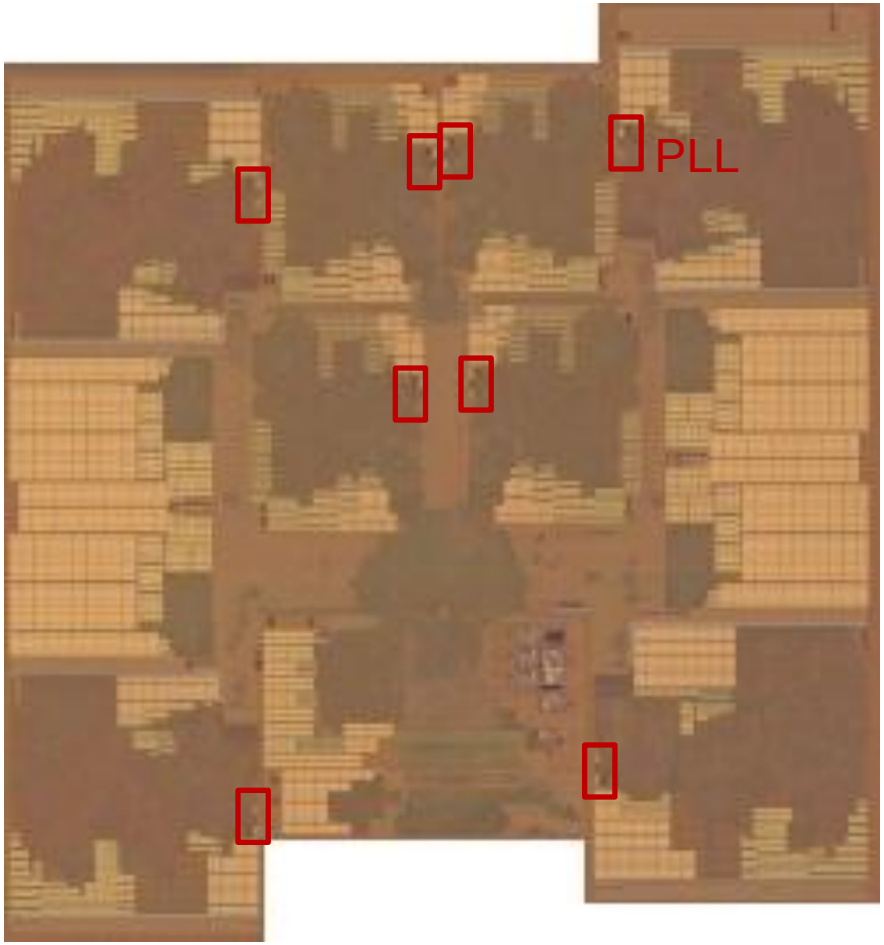
Product Examples

- NVIDIA Blackwell GPU



Product Examples

- MediaTek Dimensity 9300 SoC (figure shows CPU part)





PLL Basics

- A PLL takes an input reference clock and generates an output clock

$$f_{\text{out}} = f_{\text{ref}} \times N$$

- N can be an integer or a fractional number
- There is a frequency divider that generates a signal with

$$f_{\text{div}} = f_{\text{out}}/N$$

- After the PLL is locked, $f_{\text{ref}} = f_{\text{div}}$, and their phases are synchronized

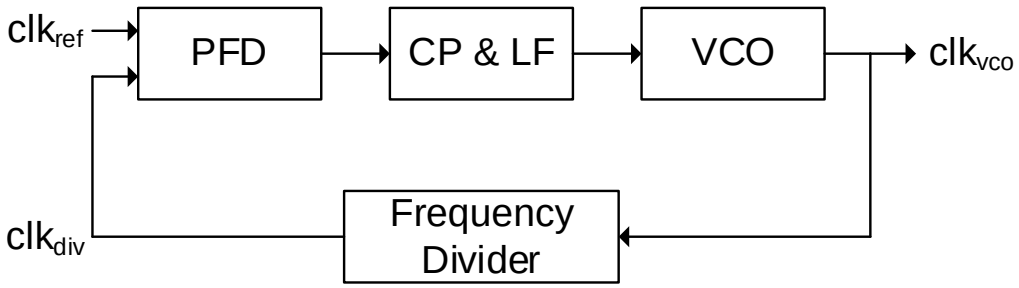
$$\Rightarrow f_{\text{out}} = f_{\text{div}} \times N$$

- Frequency ranges for clock generation
  - Reference frequency: 10 MHz – 100 MHz
  - Output frequency: 100 MHz – 5 GHz



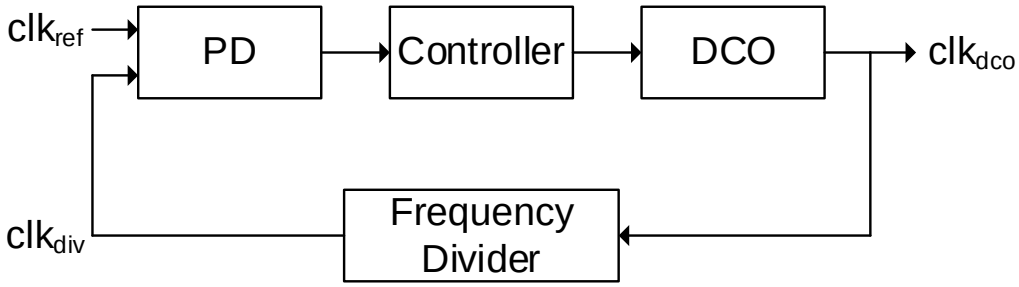
PLL Types

- Analog PLL



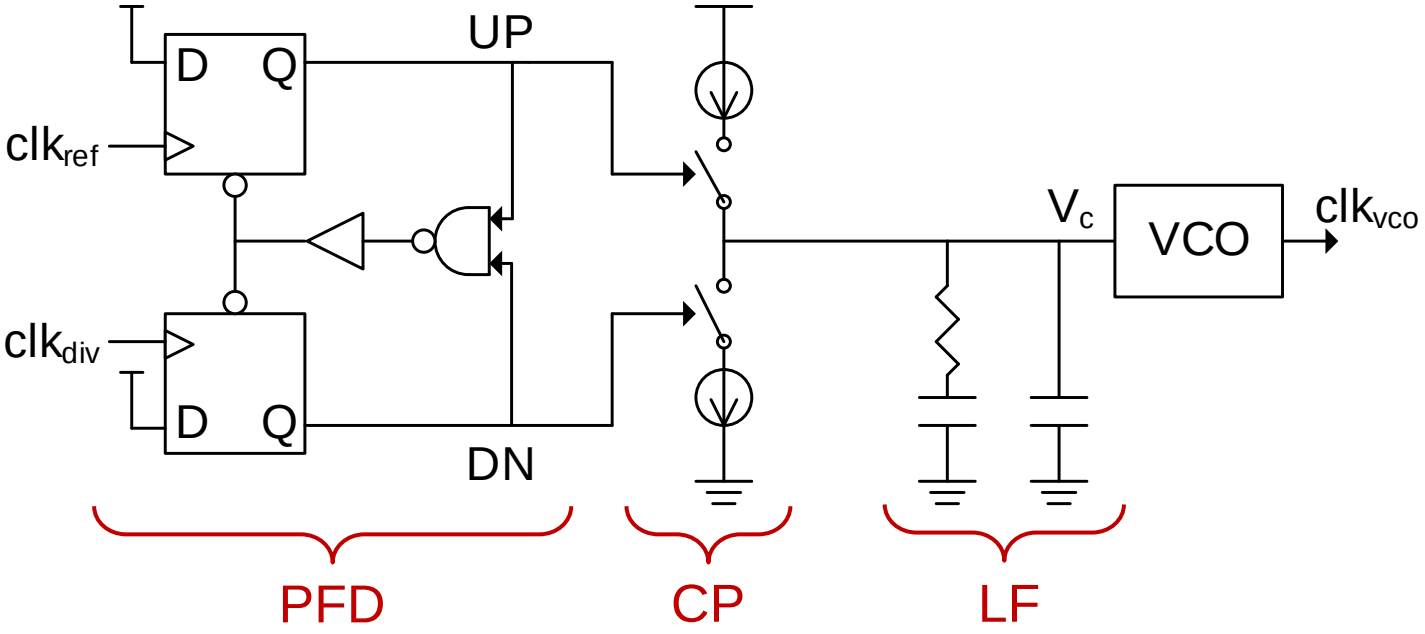
PFD: phase frequency detector
CP: charge pump
LF: loop filter
VCO: voltage controlled oscillator

- Digital PLL



PD: phase detector
Controller: digital circuit
DCO: digitally controlled oscillator

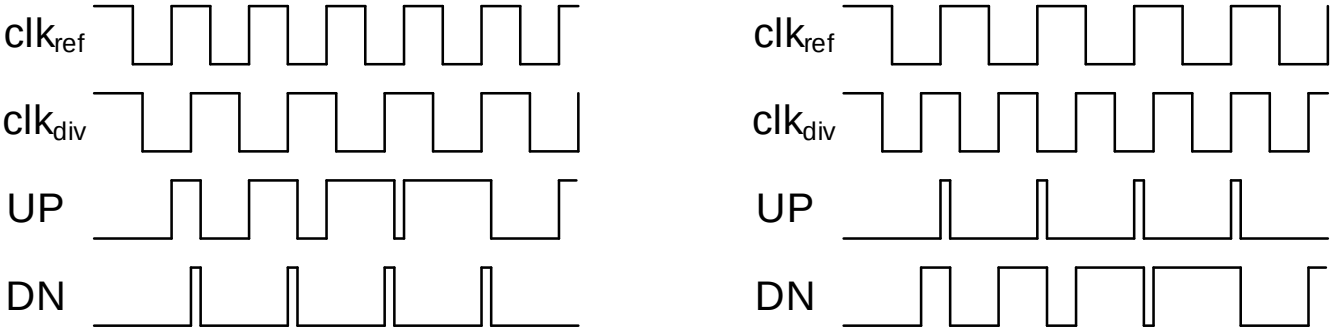
Analog PLL



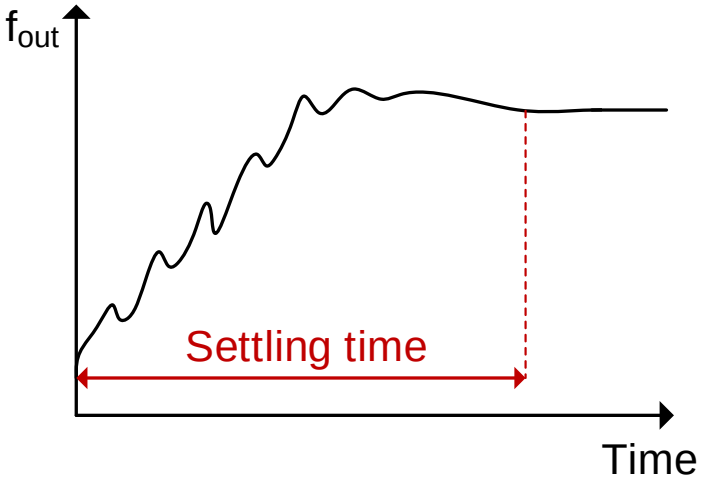
- UP = 1, DN = 0: Vc↑. UP = 0, DN = 1: Vc↓
- UP = DN = 1: reset flip-flops, Vc —. UP = DN = 0: Vc —
- The loop filter smooths Vc

Analog PLL Operation

- PFD

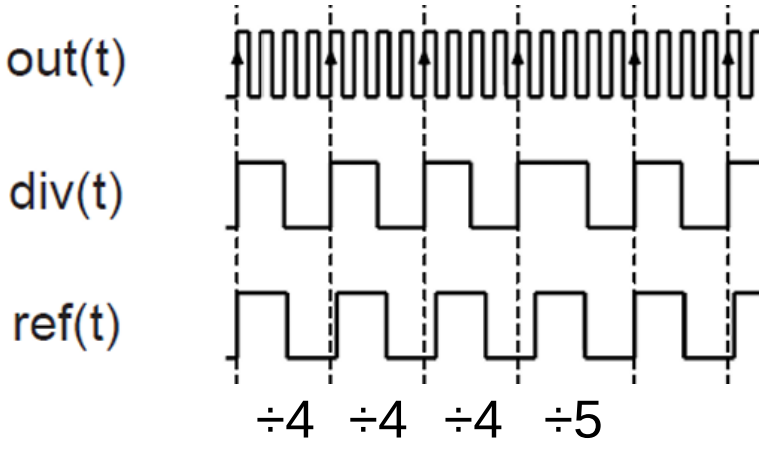


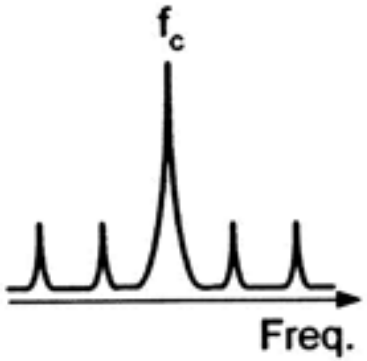
- VCO
  - Settling time (lock time)
  - 1 μs – 10 μs



Fractional-N Frequency Divider

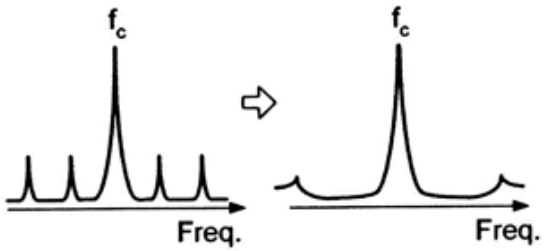
- For example, we want to divide the frequency by 4.25
- Intuitive implementation
  - Multi-modulus divider (MMD) of dividing by 4 or 5
  - Division by 5 is performed every three division by 4
  - The average frequency is divided by (4 + 4 + 4 + 5) / 4 = 4.25
- The above method creates harmonic tones (fractional spurs)





Fractional Spur Suppression

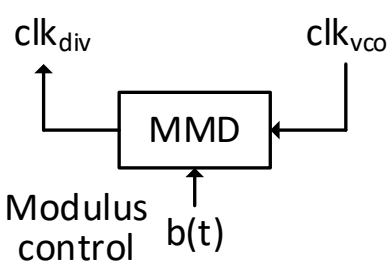
- The fractional spurs can be suppressed by randomization and noise shaping

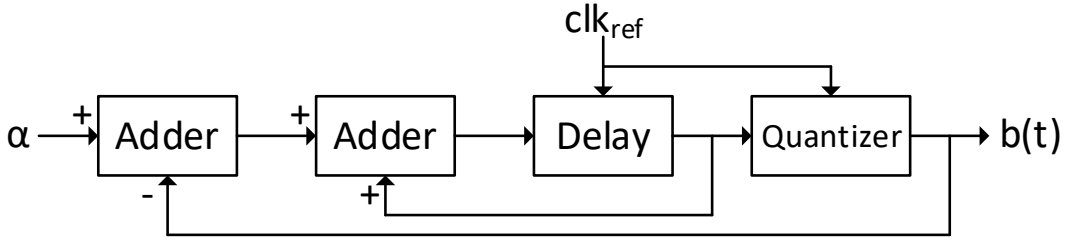


- Randomization
  - $b(t)$ is a random select signal such that its average value is α
- Noise shaping (Delta-sigma modulator)

N = integer + α

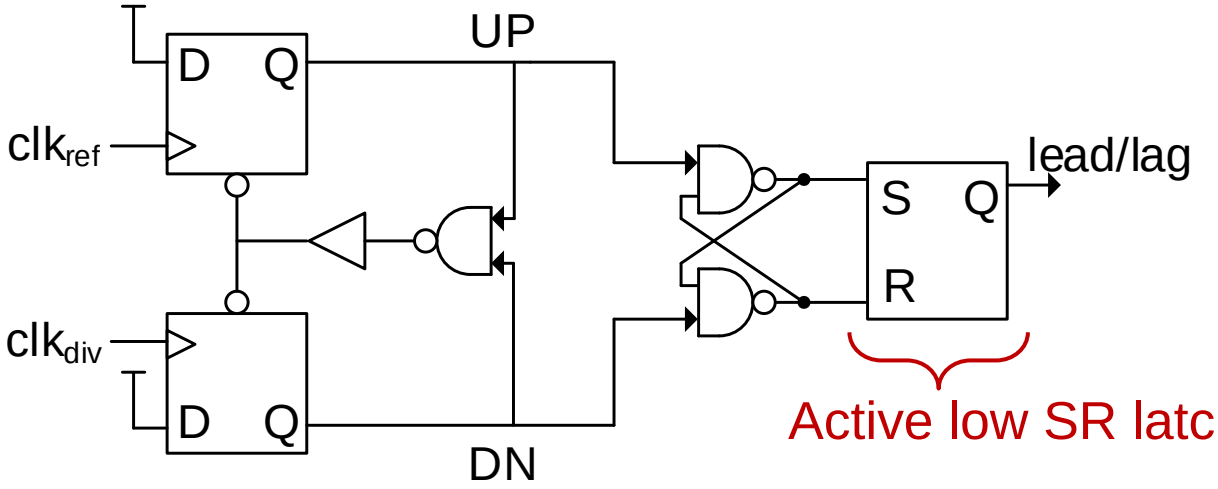
  - α is represented in finite number of bits
  - The quantizer outputs 0 or 1





Digital PLL

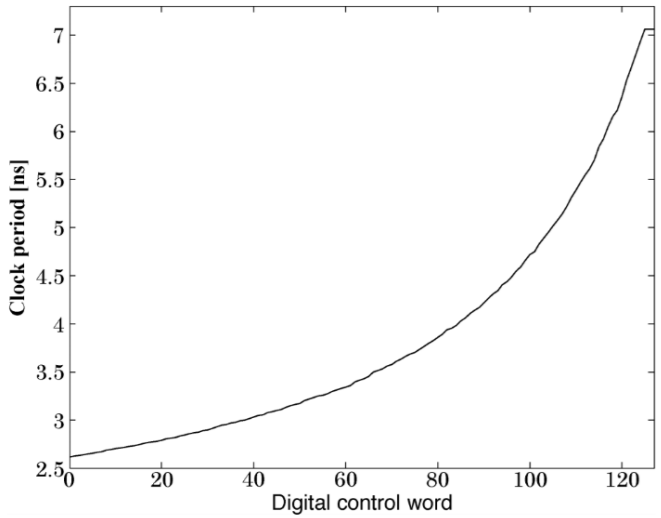
- Phase detector

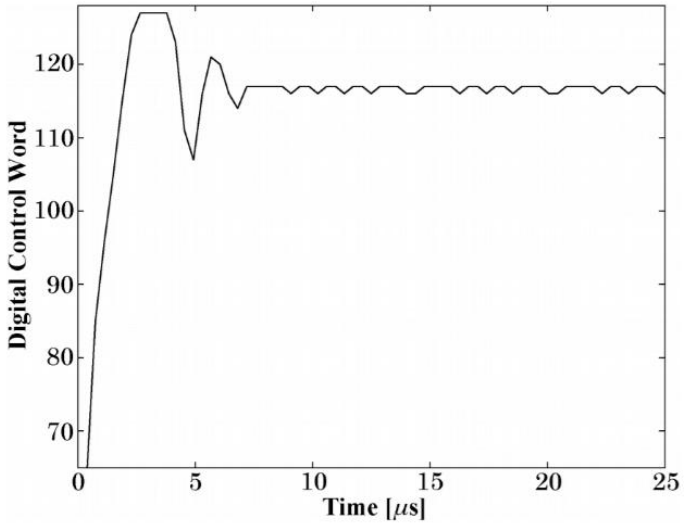


- UP = 1, DN = 0: S = 0, R = 1, lead/lag = 1. clk_{ref} leads clk_{div}
- UP = 0, DN = 1: S = 1, R = 0, lead/lag = 0. clk_{ref} lags clk_{div}
- The controller utilizes the lead/lag signal to control the DCO through certain locking algorithms

Digital PLL Operation

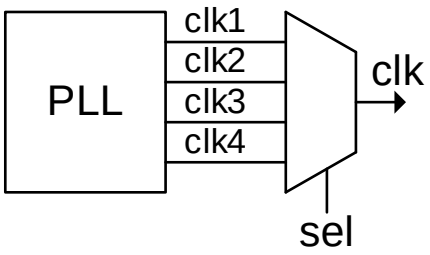
- The controller performs the locking procedure
  1. Finding frequency range
  2. Frequency acquisition
  3. Phase locking
  4. Phase tracking
- DCO



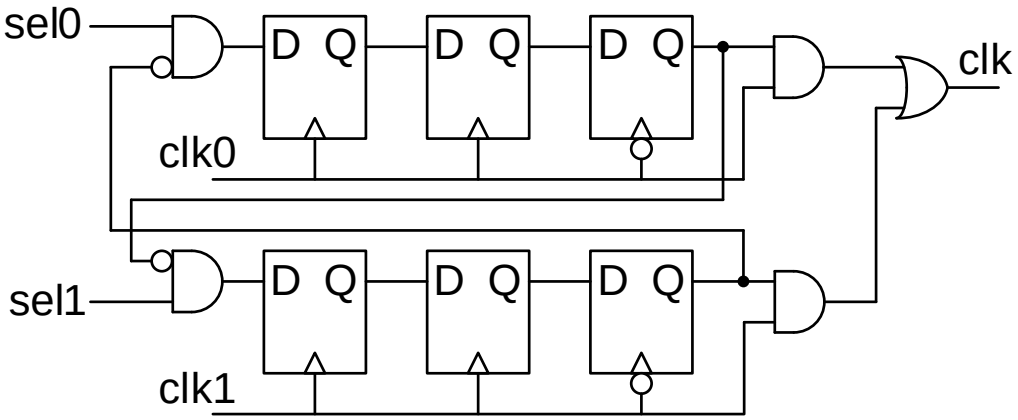


Clock Multiplexing

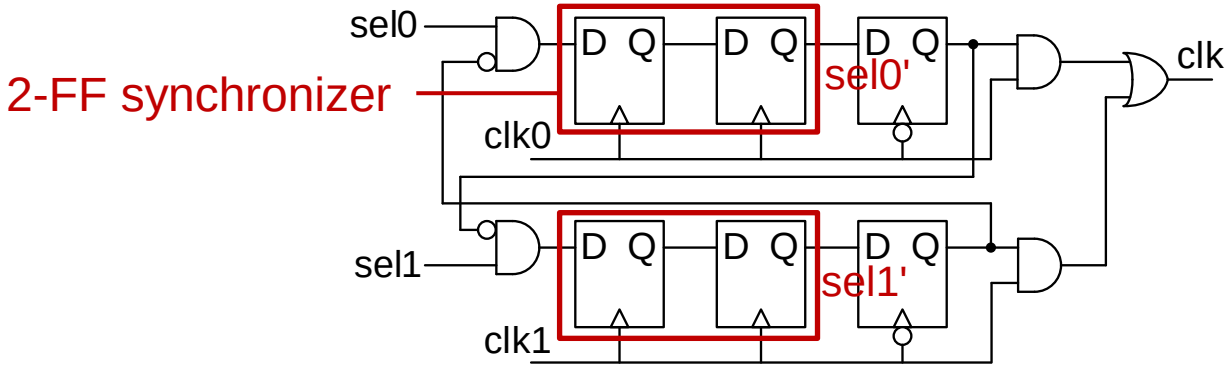
- Conventional multiplexers can introduce glitches which create functional problems

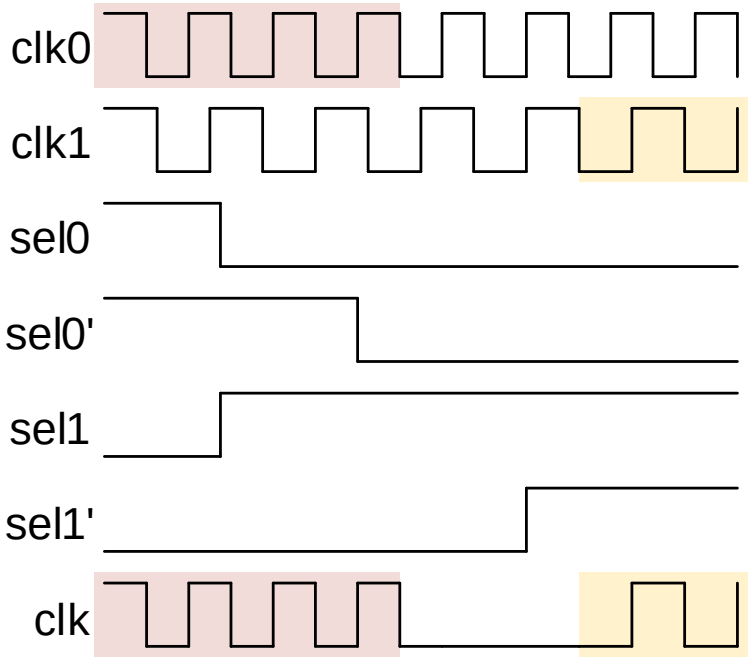


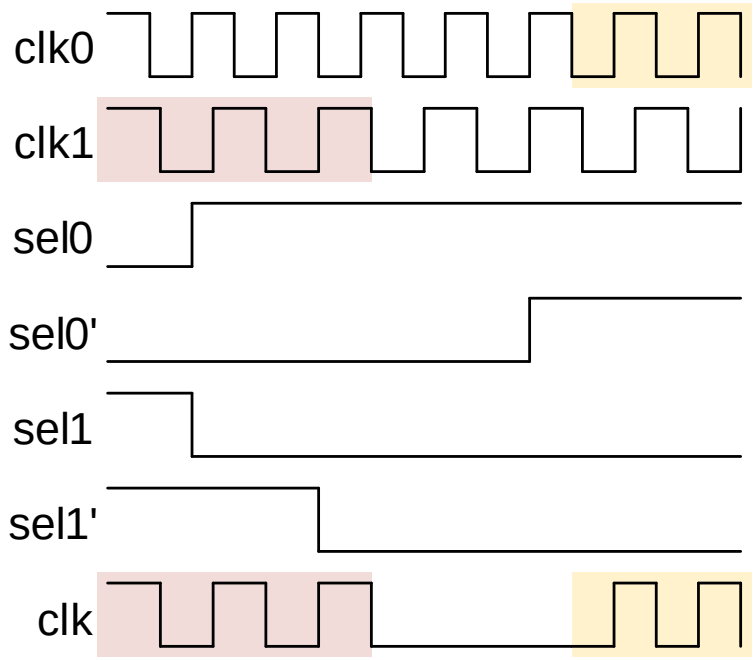
- Glitch-free clock multiplexer



Clock Multiplexing

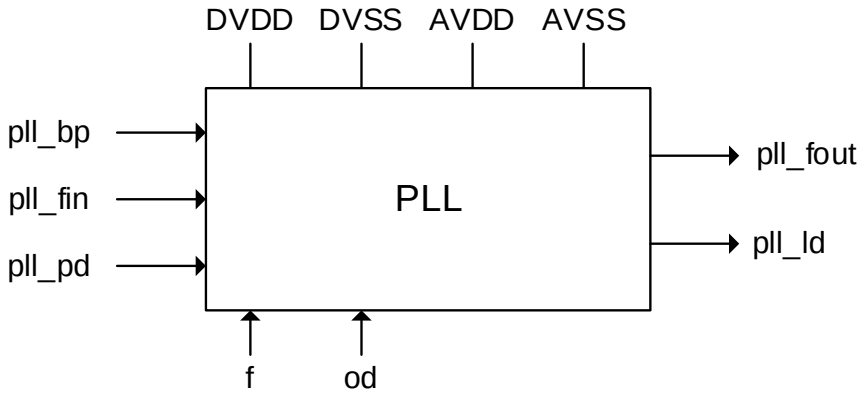






PLL Macro

- DVDD, DVSS: digital power
- AVDD, AVSS: analog power
- pll_bp: bypass
- pll_fin: input clock
- pll_pd: power down
- f: multiplier
- od: output divider



- pll_fout: output clock
- pll_ld: lock



Power-on of SoC

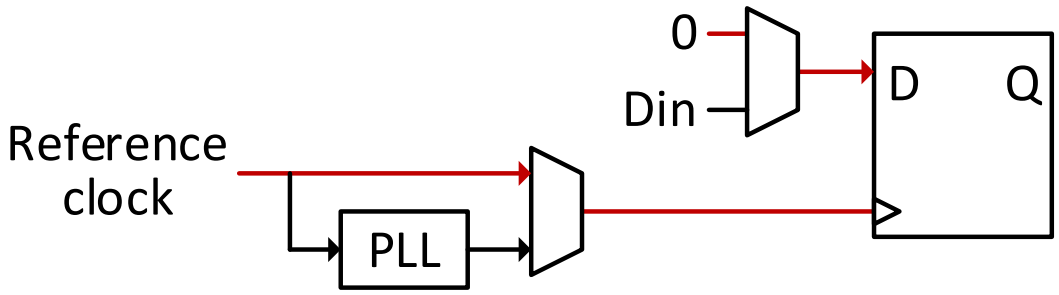
Power-On Sequence

SoC Boot-up Power Sequence — phases 0–9

Phase 0 — AON & PMU	Always-On domain, POR timer, boot straps	POR
Phase 1 — Reset asserted	POR / nRESET / DBG_RESET Hold cores	
Phase 2 — Boot ROM power	Boot ROM + minimal SRAM Fuse controller	
Phase 3 — First instruction	Boot core fetches first instruction	
Phase 4 — Clock bring-up	XO → Program PLLs → Wait for LOCK	PLL_LOCK
Phase 5 — Core domains	Rail → Isolation → Clock → Reset	
Phase 6 — Security	Read fuses, RoT, authenticate SBL	
Phase 7 — DDR training	Power rails, PHY, Read/write leveling	DDR_INIT_DONE
Phase 8 — Hand-off	Boot ROM → SBL / UEFI / U-Boot Enable GPU / ISP	
Phase 9 — OS PM	DVFS, runtime gating, Suspend / Resume	

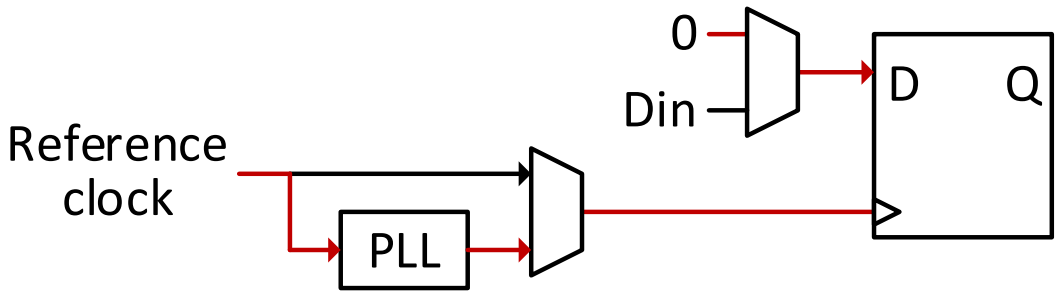
Power-On Reset

- The power-on reset (POR) circuit detects whether the power supply is ready
- It asserts reset signals to all important flip-flops
- Note that the PLL has not locked in this stage
- The flip-flops are reset with the aid of the reference clock



Clock Generation

- After POR, the PLL starts to generate system clock
- PLL indicates its state by PLL_LOCK signal
- After PLL_LOCK = 1, the system clock is prepared
- The clock source of flip-flops is switched to the system clock
  - The clock MUX is used



Reset Release

- The POR circuit deasserts reset signals after the PLL is locked and the clock source is switched to system clock
- The reset signal from POR circuit is asynchronous to the system clock
  - The two flip-flop synchronizer is used to reduce metastability and align reset signals

